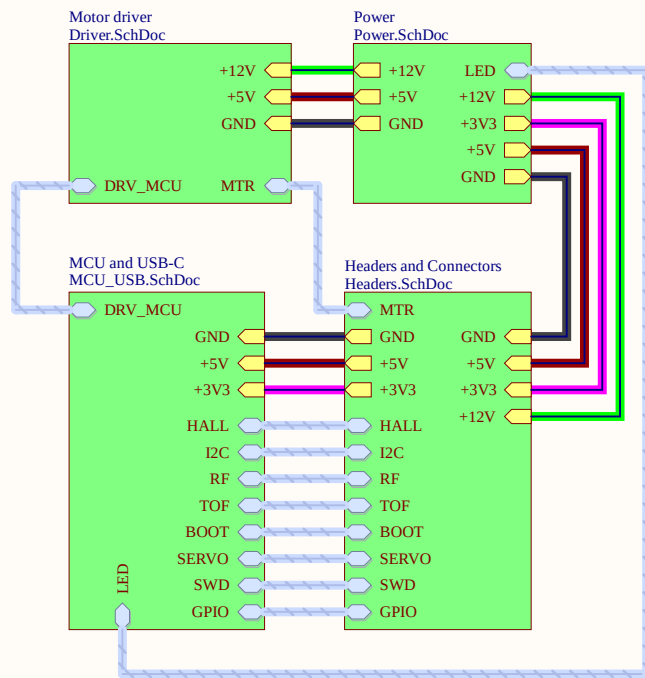
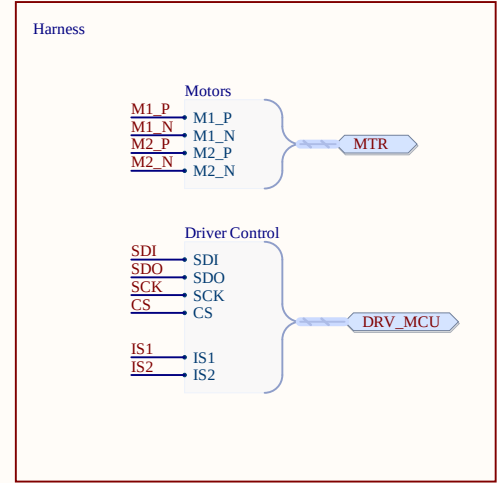
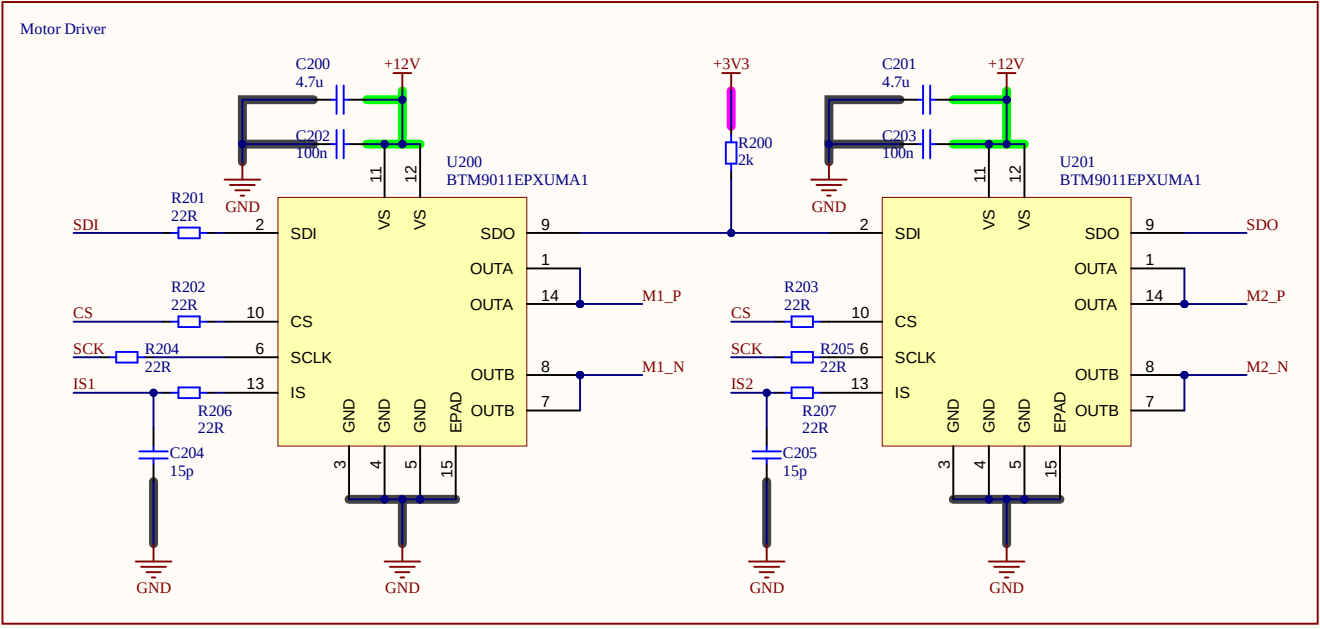
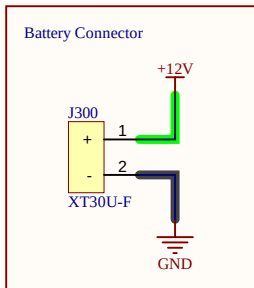
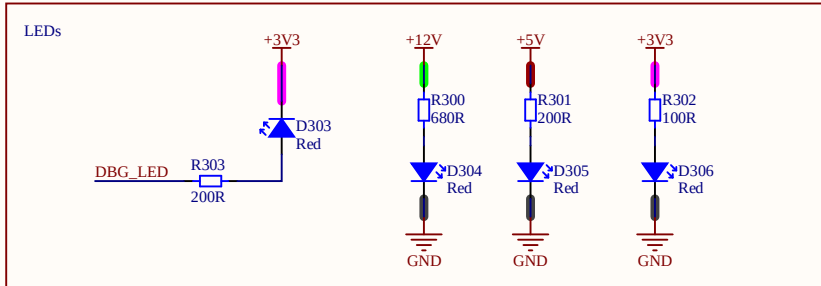
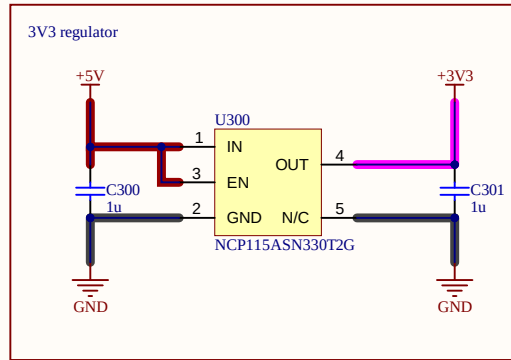
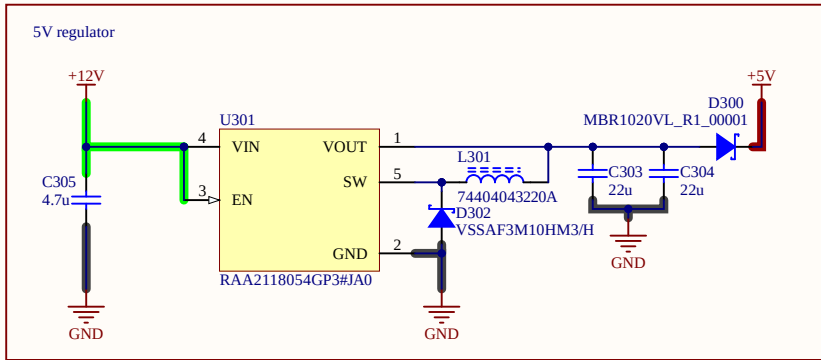
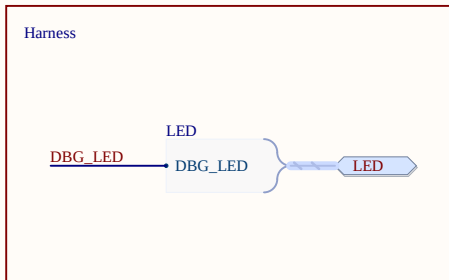




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Date:	3/09/2026	Sheet of
File:	C:\Users\...\Overview.SchDoc	Drawn By:
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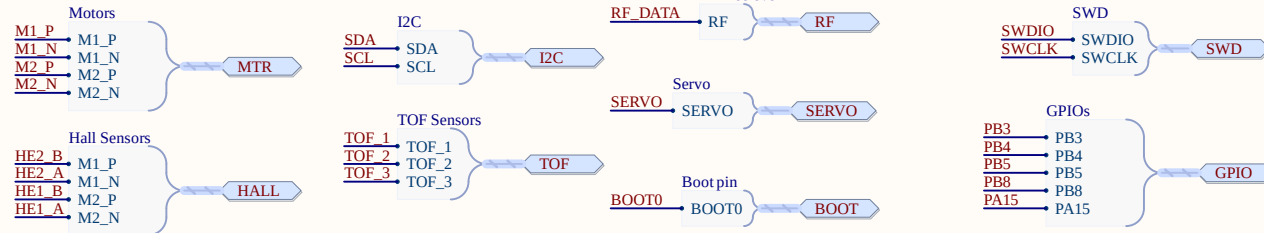


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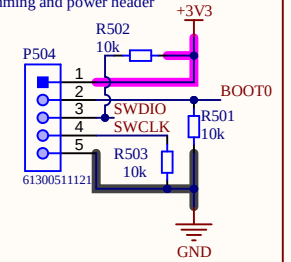


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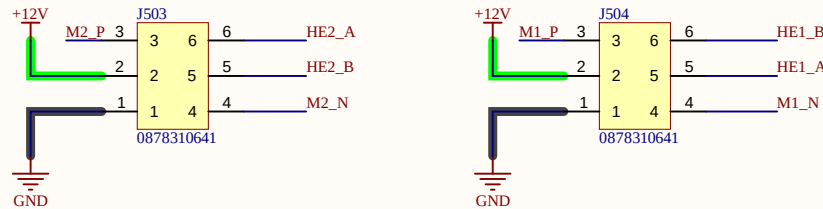
Harness



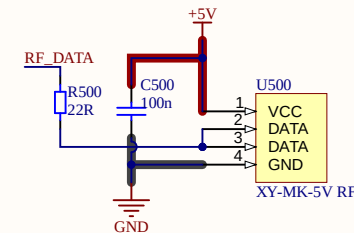
Boot, programming and power header



Motor Connectors

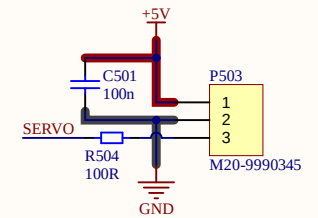


Receiver



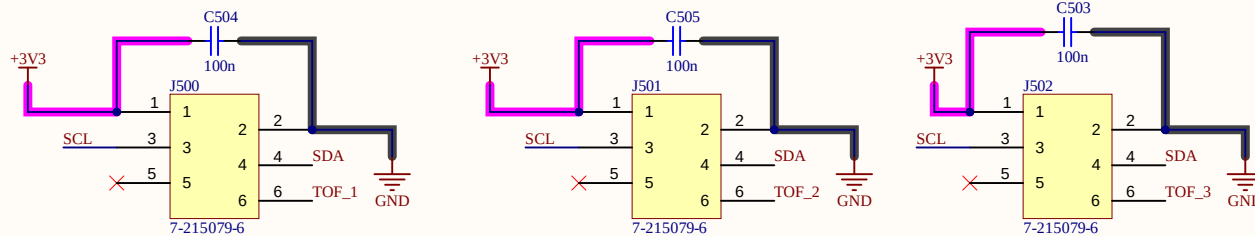
Receiver boards
solderable header

Servo



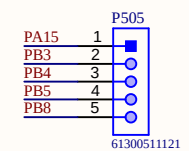
Servo motor
header

TOF Sensors



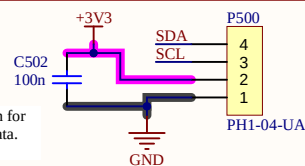
IDC connectors for TOF sensors.
2.54mm open headers connection
is unreliable

Extra pins



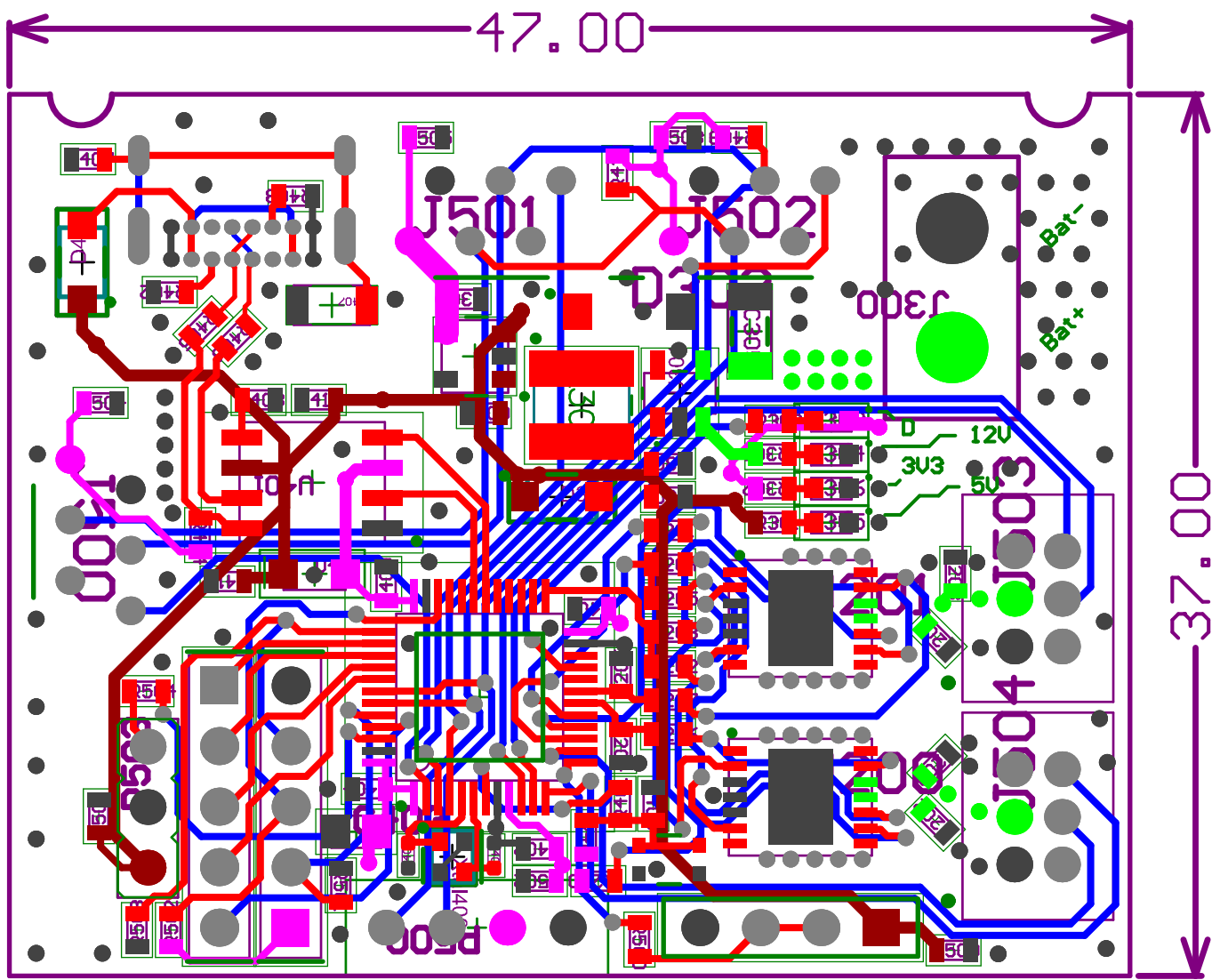
Extra header for
future use

OLED



OLED screen for
displaying data.

Title		
Size A3	Number	Revision
Date: 3/09/2026	Sheet of	
File: C:\Users\...\Headers.SchDoc	Drawn By:	



Comment	Description	Designator	Footprint	LibRef	Quantity
4.7u	MLCC, 0603, 4.7u, 10%, 35V, XSR	C200, C201	MLCC_0603_M	CMP-001-1257-1	2
100n	MLCC, 0603, 100n, 10%, 35V, XSR, MLCC, 0603, 100n, 10%, 50V, XSR	C202, C203, C400, C401, C402, C404, C406, C411, C500, C501, C502, C503, C504, C505	MLCC_0603_M	CMP-001-0755-1, CMP-001-0815-1	14
15p	MLCC, 0603, 15p, 5%, 50V, COG/NPO	C204, C205	MLCC_0603_M	CMP-001-0474-1	2
1u	MLCC, 0603, 1u, 10%, 50V, XSR	C300, C301, C408, C410	MLCC_0603_M	CMP-001-0565-1	4
22u	MLCC, 0603, 22u, 10%, 16V, XSR	C303, C304	MLCC_0603_M	CMP-001-1264-1	2
4.7u	MLCC, 1206, 4.7u, 10%, 35V, XSR	C305	MLCC_1206_M	CMP-001-1124-2	1
4.7u	MLCC, 0805, 4.7u, 20%, 50V, XSR	C403	MLCC_0805_M	CMP-001-1830-1	1
15p	MLCC, 0402, 15p, 10%, 25V, COG/NPO	C405, C406	MLCC_0402_M	CMP-001-1853-1	2
1n	MLCC, 0603, 1n, 10%, 50V, XSR	C407	MLCC_0603_M	CMP-001-0547-1	1
MFR020M_R1_00001	Rectifier Diode, Schottky, 1 Phase, 1 Element, 1A, 20V V(RRM), Silicon	D300, D400	FP: MFR020M_R1_00001-MFG	CMP-ROOT-01310-1	2
V5S4F3M10HM3H	Rectifier Diode, Schottky, 1 Phase, 1 Element, 2.3A, 100V V(RRM), Silicon, DO-221AC	D302	V5S4F3M10HM3H-Footprint-1	CMP-ROOT-01308-1	1
Red	Generic red 0603 LED	D303, D304, D305, D306	0603 RED LED	CMP-ROOT-00025-2	4
1N5817G	Rectifier Diode, Schottky, 1 Phase, 1 Element, 1A, 20V V(RRM), Silicon, DO-204AL	D401	SOD-323 based on 1N5817WS	CMP-ROOT-01323-1	1
X130U-F	Power Connector, USB Connector, 16 Contact(s), Female, Right Angle, Solder Terminal, Locking, Receptacle	J000	X130U-F	CMP-ROOT-01262-3	1
USB4085-GF-A	USB Connector, 16 Contact(s), Female, Right Angle, Solder Terminal, Locking, Receptacle	J000	FP: USB4085-GF-A-MFG	CMP-001-1800-1	1
7-215079-6	Conn FFC Connector P 6 POS 1.27mm Solder ST Thru-Hole Automotive Box/Cartron	J500, J501, J502	7-215079-6-Footprint-1	CMP-ROOT-01311-2	3
0878310641	Conn Shrouded Header (2 Sides) HDR 6 POS2mm Solder ST Top Entry Thru-Hole Mini-Cold Tube	J503, J504	0878310641	CMP-ROOT-01318-3	2
74404043220A	General Purpose Inductor, 22uH, 20%, 1 Element, Nickel-zinc Ferrite-Core, SMD, 1636	L301	ME-LQS_4025	CMP-ROOT-01309-1	1
PH2-04-4A	4 POS2.54mm Vertical THT Header	P500	4Pos Pin Header Vertical	CMP-002-000004-1	1
M20-9990345	Connector Header Through Hole 3 position 0.100" (2.54mm)	P503	3Pos Pin Header	CMP-ROOT-00217-1	1
61300511121	Conn Unshrouded Header HDR5 POS 2.54mm Solder ST Top Entry Thru-Hole Bag	P504, P505	61300511121	CMP-ROOT-01020-1	2
2k	Resistor, 0603, 2k, 1%, 1/10W	R000	Resistor_0603_M	CMP-001-0084-2	1
22R	Resistor, 0603, 22R, 1%, 1/10W	R001, R002, R003, R004, R005, R006, R007, R008, R009, R000	Resistor_0603_M	CMP-001-0038-3	10
680R	Resistor, 0603, 680R, 1%, 1/10W	R000	Resistor_0603_M	CMP-001-0074-3	1
200R	Resistor, 0603, 200R, 1%, 1/10W	R001, R003	Resistor_0603_M	CMP-001-0060-2	2
100R	Resistor, 0603, 100R, <5%, <1/16W, Resistor, 0603, 100R, 1%, 1/10W	R002, R401, R504	Resistor_0603_M	CMP-0001-00252-1, CMP-001-0054-5	3
10k	Resistor, 0603, 10k, 1%, 1/10W, Resistor, 0603, 10k, <5%, <1/16W	R000, R501, R502, R503	Resistor_0603_M	CMP-001-0101-2, CMP-0001-00257-1	4
5.1k	Resistor, 0603, 5.1k, 1%, 1/10W	R402, R403	Resistor_0603_M	CMP-001-0094-2	2
1.5k	Resistor, 0603, 1.5k, 1%, 1/10W	R404	Resistor_0603_M	CMP-001-0081-2	1
1M	Resistor, 1206, 1M, 1%, 1/4W	R407	Resistor_1206_M	CMP-0001-00051-1	1
4.7k	Resistor, 0603, 4.7k, 1%, 1/10W	R409, R410	Resistor_0603_M	CMP-001-0093-2	2
USLPT2819D14TR	Keypad Switch, SPST, 0.05A, 1.98N, Solder Terminal, Panel Mount	S400	USLPT2819D14TR-Footprint-1	CMP-ROOT-01290-1	1
BTM6011EPJUMA1	D.C.MV Drive, Brush, F500CEC, TSSOPEP, 14 Pins Compliant: Yes	U200, U201	BTM6011EPJUMA1	CMP-00022-3	2
NCP115A9N030T2G	LED Regulator, 300 mA, Low Iq, High PSRR, High Accuracy 3.3V, TSO-P, Active Discharge, Normal Vout Sleep/Noe	U300	TSCP-5	CMP-00026-1	1
PAW2118054G3HJ40	Switching Regulator, Current-mode, 0.3A, 2.28A Sw Switching Freq Max, PDSOS	U301	P5_0648_MFG	CMP-00019-3	1
STM32F103C8T6	MCU 32-bit STM32F1 ARM Cortex-M3 RISC, 64Kb Flash 2.5V/3.3V 48-Pin LQFP Tray	U400	STM-LQFP48_M	CMP-00012-3	1
HT42BS34-2	USB to UART 3Mbps USB 2.0 SCD-8 Interface Controllers, Pkts	U401	SCP-8 150mI	CMP-00027-2	1
XY-MK-SV-RF	433MHz receiver module, 5V tolerant	U500	2.54mm4 POS XY-MK-SV-RF	CMP-00015-1	1
ABM11-16.000M42-1-NIG-T	Crystal 16MHz ±10ppm (Typ) ±15ppm (Stability) 12pF FLN D 150OHm 4-Pin Ultra Mini-CSMD TR	X4400	FP-ABM11-MFG	CMP-ROOT-01254-3	1